



ICAIN-2026
International Conference on Artificial
Intelligence and Networking

26th -28th November 2026
Organized by
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In association with
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******* CALL FOR PAPERS *******

SPECIAL SESSION ON

Robust and Edge-Intelligent AI for Wearable Biomedical Signal Analytics and Digital Health

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SESSION DESCRIPTION:

Wearable sensing and remote monitoring technologies have changed the way in which we can provide care to individuals by allowing us to continuously monitor and acquire physiological data from the body through wearable devices. The use of wearable devices allows for continuous, non-invasive acquisition of ECG, PPG, EEG, EMG, respiratory data and inertial signals. Although wearable technology has allowed for the opportunity to continuously monitor a patient's physiological signals, the ability to derive reliable clinical insights continues to be difficult due to motion artifact, noise, sensor drift, variability between subjects, limited amounts of labelled training data, variability in wearable technology used, and strict latency/power constraints placed on wearable devices. This special session is focused on developing the use of artificial intelligence that is both robust and deployable for wearable biomedical signal analysis. This will be achieved by using principled signal processing methods combined with modern machine learning/distributed learning techniques (for example, transformers, self-supervised learning, multimodal fusion, and uncertainty-aware decision making). Contributions addressing topics such as TinyML/edge inference, federated learning for privacy preservation, fairness for different demographic groups, explainability and clinically relevant evaluation methodologies for validating digital health and telemedicine applications are encouraged.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- Robust artifact removal and signal quality indexing (SQI) for ECG/PPG/EEG/EMG
- Time-frequency and nonlinear/entropy measures for wearable biosignals
- Deep learning for 1D physiological time-series (CNN/RNN/TCN/Transformers)
- Self-supervised / semi-supervised / label-efficient learning for biosignals
- Multimodal fusion: biosignals + IMU + audio + camera-based rPPG
- Domain adaptation/generalization across subjects/devices/environments
- Uncertainty estimation, calibration, risk-aware decision making
- Explainable AI for biosignals (attention/saliency/rule-based)
- Edge AI/TinyML: compression, quantization, pruning, streaming inference
- Federated learning, privacy/security, fairness for digital health/loHT

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit papers for this particular theme session on **Robust and Edge-Intelligent AI for Wearable Biomedical Signal Analytics and Digital Health** *on or before* [30th June 2026]. All submissions must be original and may not be under review by another publication. INTERESTED AUTHORS SHOULD CONSULT THE CONFERENCE'S GUIDELINES FOR MANUSCRIPT SUBMISSIONS at <https://www.icain-conf.com/downloads>. All submitted papers will be reviewed on a double-blind, peer-review basis.

NOTE: While submitting a paper in this special session, please specify **Robust and Edge-Intelligent AI for Wearable Biomedical Signal Analytics and Digital Health** at the top (above paper title) of the first page of your paper.

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